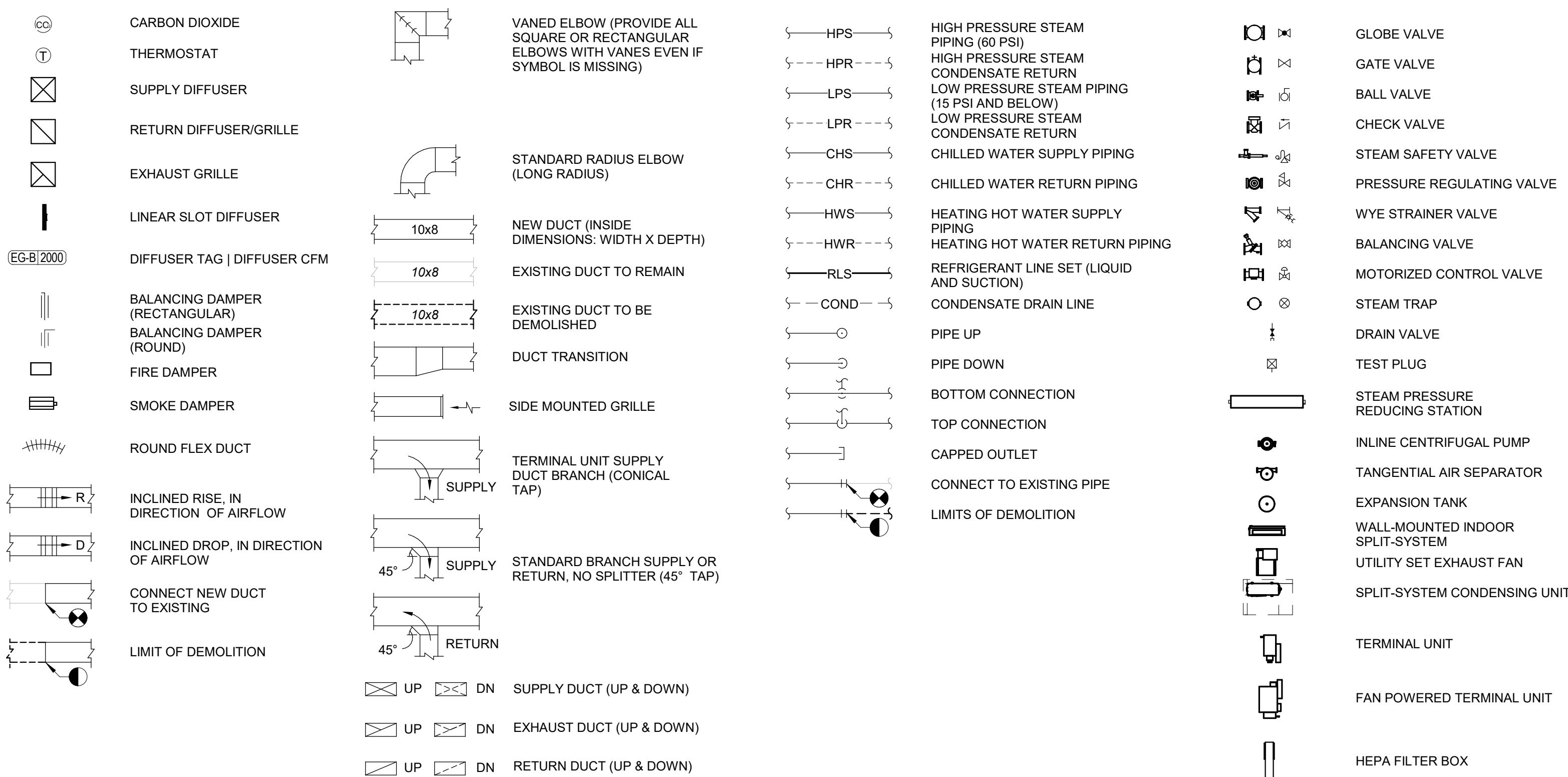


MECHANICAL ABBREVIATIONS

A/E	ARCHITECT / ENGINEER	DPA	DIFFERENTIAL PRESSURE ASSEMBLY	IN	INCHES	R/E	RETURN OR EXHAUST
AAHX	AIR TO AIR HEAT EXCHANGER	DPS	DIFFERENTIAL PRESSURE SENSOR	IN HG	INCHES OF MERCURY	RA	RETURN AIR
AB	AIR BLENDER	DX	DIRECT EXPANSION	IN WC	INCH WATER COLUMN	RAT	RETURN AIR TEMPERATURE
AAV	AUTOMATIC AIR VENT	DXCC	DIRECT EXPANSION COOLING COIL	IN WG	INCH WATER GAUGE	RDS	ROOM DATA SHEETS
ACC	AIR-COOLED CONDENSER			IN-LB	INCH-POUND	REA	RELIEF AIR
ACCH	AIR-COOLED CHILLER	EA	EXHAUST AIR	IPLV	INTERGRATED PART LOAD VALUE	RF	RETURN FAN
ACCU	AIR-COOLED CONDENSING UNIT	EAT	ENTERING AIR TEMPERATURE	IRH	INTRARED HEATER	RG	RETURN GRILLE
ACCU	AIR-CONDITIONING UNIT	ECC	ENGINEERING CONTROL CENTER	IS	INSET SCREEN	RH	RELATIVE HUMIDITY
AED	AUTOMATIC CONTROL DAMPER, MODULATING	EER	ENERGY EFFICIENCY RATIO	IV	INLET VANES	RHC	REHEAT COIL
ACD-TP	AUTOMATIC CONTROL DAMPER, TWO POSITION	EG	EXHAUST GRILLE	kg	KILOGRAM	RL	REFRIGERANT LIQUID LINE
AD	ACCESS DOOR	EGS	EMERGENCY GAS SHUTOFF	kg/HR	KILOGRAM PER HOUR	RSL	RUN-LOAD AMPERE
AF	AFTER FILTER	EH	EXHAUST HOOD	KW	KILOWATT	RPM	REVOLUTIONS PER MINUTE
AFCV	AIR FLOW CONTROL VALVE	EJ	EXPANSION JOINT	KWh	KILOWATT HOUR	RS	REFRIGERANT SUCTION
AFMD	AIR FLOW MEASURING DEVICE	ENT	ENTERING			RTU	ROOF TOP UNIT
AFW	AIR FOIL WHEEL (FAN)	ESP	EXTERNAL STATIC PRESSURE	L	LITER	RV	RELIEF VALVE
AHU	AIR-HANDLING UNIT	EWT	ENTERING WATER TEMPERATURE	L/s	LITERS PER SECOND		
AMP	AMPERES	EX	EXISTING	LAT	LEAVING AIR TEMPERATURE	SAD	SOUND ATTENUATING DEVICE
AP	ACCESS PANEL	F	FAHRENHEIT	LBS/HR	POUNDS PER HOUR	SAT	SUPPLY AIR TEMPERATURE
APD	AIR PRESSURE DROP	FAT	FLOAT AND THERMOSTATIC	LF	LINEAR FOOT (FEET)	SC	SHADING COEFFICIENT
ARI	AIR CONDITIONING AND REFRIGERATION INSTITUTE	F/SDPR	FAN COIL UNIT (4 PIPE)	LH	LATENT HEAT	SCFM	STANDARD CUBIC FEET PER MINUTE
AS	AIR SEPARATOR	FA	FREE AREA	LPR	LOW PRESSURE RETURN (STEAM CONDENSATE)	SD	SMOKE DETECTOR
ASME	AMERICAN SOCIETY OF MECHANICAL ENGINEERS	FC	FLEXIBLE CONNECTION	LPS	LOW PRESSURE SUPPLY (STEAM)	SDR	SUPPLY AIR DIFFUSER
AW	AIR WASHER	FCU	FAN COIL UNIT (4 PIPE)	LSD	LINEAR SLOT DIFFUSER	SJ	SMOKE DAMPER
AXF	AXIAL FLOW	FCW	FORWARD CURVED WHEEL (FAN)	LTCP	LOCAL TEMPERATURE CONTROL	SJR	SMOKE DAMPER (RETURN)
		FD	FLOOR DRAIN	LTVG	LOCAL TEMPERATURE CONTROL	SNS	SMOKE DAMPER (SUPPLY)
		FE	FIRE DAMPER	LVL	LEAVING	SEN	SENSIBLE HEAT
		FF	FINAL FILTER	LVT	LOW VENT	SH	STEAM HUMIDIFIER
B	BOILER	FM	FLOW METER	LWT	LEAVING WATER TEMPERATURE	SHC	STEAM HEATING COIL
BD	BUTTERFLY DAMPER	FPM	FEET PER MINUTE			SI	SQUARE INCHES
BDD	BACKDRAFT DAMPER	FPS	FEET PER SECOND	M	METER, SI UNIT	SP	STATIC PRESSURE
BDR	BASE BOARD RADIATOR	FPR	FIBER REINFORCED POLYESTER	M/s	METERS PER SECOND (OR METERS/SECOND)	SP GR	SPECIFIC GRAVITY
BFP	BACKFLOW PREVENTER	FS	FLOW SWITCH	MA	MIXED AIR	SPD	SUPPLY PROCESS AND DISTRIBUTION
BFT	BOILER PLANT FIRE TUBE	FSSTAT	FREESTAT	MAT	MIXED AIR TEMPERATURE	SPRV	STEAM PRESSURE REDUCING VALVE
BHP	BRAKE HORSEPOWER	FT	FEET	MB	MANUAL AIR VENT	SPS	STATIC PRESSURE SENSOR
BHW	HOT WATER HEATING BOILER	FT-LB	FOOT-POUND	MB	MAXIMUM	SQ FT	SQUARE FOOT (FEET)
BHX	BOILER BLOWDOWN HEAT EXCHANGER	FTR	FIN TUBE RADIATION	MAX	MAXIMUM	SR	SUPPLY AIR REGISTER
BW	BACKWARD INCLINED WHEEL (FAN)	FV	FACE VELOCITY	MB	MIXING BOX	SS	STAINLESS STEEL
BSC	BIOLOGICAL SAFETY CABINETS			MBH	1000 BTUH	SSR	SOLID SEPARATOR
BT	BUFFER TANK	GA	GAUGE	MCA	MINIMUM BRANCH CIRCUIT AMPACITY	ST	STEAM TRAP
BTU	BRITISH THERMAL UNIT	GAL	GALLONS	MER	MECHANICAL EQUIPMENT ROOM	SV	STEAM SAFETY VALVE
BTUH	BRITISH THERMAL UNIT PER HOUR	GH	GRAVITY HOOD	MERV	MECHANICAL EQUIPMENT ROOM	SVS	STEAM VENT SILENCER
		GPD	GALLONS PER DAY	VALUE		T & PCV	TEMPERATURE AND PRESSURE CONTROL VALVE
C	CENTIGRADE (CELCIUS)	GPH	GALLONS PER HOUR	MHP	MOTOR HORSEPOWER	TAB	TESTING, ADJUSTING, BALANCE
CC	COOLING COIL	GPM	GALLONS PER MINUTE	MIN	MINIMUM	TDH	TEMPERATURE DIFFERENCE
CCD	COOLING COIL CONDENSATE DRAIN	GS	GALVANIZED STEEL	MM	MILLIMETER	TDS	TOTAL DISSOLVED SOLIDS
CENT	CENTIMETER			MOV	MOTOR OPERATED VALVE	TRAP	TRAP
CFH	CUBIC FEET PER HOUR	H	HUMIDIFIER	MPC	MEDIUM PRESSURE RETURN (STEAM CONDENSATE)	TSP	TOTAL STATIC PRESSURE
CFM	CUBIC FEET PER MINUTE	H&CW	HOT & COLD WATER	MPS	MEDIUM PRESSURE SUPPLY (STEAM)	TSTAT	THERMOSTAT
CFI	CUBIC FEET	H&C	HOUSEKEEPING AID CLOSET	MTD	MEAN TEMPERATURE DIFFERENCE	TU	TERMINAL UNIT
CFP	CHEMICAL FEED PUMP	HB	HOSE BIBB	MVD	MANUAL VOLUME DAMPER		
CH	CHILLER	HC	HEATING COIL	NA	NOT APPLICABLE	UC	UNDER CUT
CHWP	CHILLED WATER PUMP	HD	HEAD	NC	NOISE CRITERIA	UH	UNIT HEATER
CHW	CHILLER WATER	HO	HOOD	NO	NORMALLY CLOSED	UL	UNDERWRITERS LABORATORY
CHWR	CHILLED WATER RETURN	HHW	HEATING HOT WATER	NC	NOISE CRITERIA	V	VALVE
CHWS	CHILLED WATER SUPPLY	HCW	HEATING HOT WATER COIL	NO	NORMALLY OPEN	VAF	VANE-AXIAL FAN
CI	CAST IRON	HHWP	HEATING HOT WATER PUMP	NOA	NATIONAL OCEANIC & ATMOSPHERIC ADMINISTRATION	VAV	VARIABLE AIR VOLUME
CM	CARBON MONOXIDE	HHWR	HEATING HOT WATER RETURN	NOM	NOMINAL	VAV	VOLUME DAMPER (MANUAL OPPOSED BLADE)
CM	CUBIC METER	HHWS	HEATING HOT WATER SUPPLY	NPLV	NON-STANDARD PART LOAD VALUE	VDF	VARIABLE FREQUENCY DRIVE
CMS	CUBIC METER PER SECOND	HQA	HAND/OFF/AUTOMATIC	NPISH	NET POSITIVE SUCTION HEAD	VH	VETERANS HEALTH ADMINISTRATION
CO	CLEAN OUT	HP	HEAT PUMP	NTS	NOT TO SCALE	VI	VIBRATION ISOLATOR
CO2	CARBON DIOXIDE	HPD	HIGH PRESSURE DRAIN TRAP	OAI	OUTSIDE AIR	VP	VACUUM PUMP
COMP	COMPRESSOR UNIT	HPR	HIGH PRESSURE RETURN (STEAM CONDENSATE)	OAI	OUTSIDE AIR INTAKE	VPS	VARIABLE PRIMARY SYSTEM
COP	COEFFICIENT OF PERFORMANCE	HRC	HEAT RECOVERY COIL	OD	OUTSIDE DIAMETER	VR	VACUUM (STEAM CONDENSATE)
CP	CONDENSATE PUMP	HRC	HEAT RECOVERY COIL	OSA	OUTSIDE SUPPLY AIR (CONDITIONED)	RETURN	
CU	CONDENSING UNIT	HRO	HEAT RECOVERY ROOM	P	PUMP	VSD	VARIABLE SPEED DRIVE
CUH	CABINET UNIT HEATER	HRP	HYDRONIC RADIANT (CEILING) PANEL	PC	PUMPED CONDENSATE	VUH	VERTICAL UNIT HEATER
CV	CONSTANT VOLUME	HRW	HEAT RECOVERY WHEEL	PCF	POUNDS PER CUBIC FOOT (FEET)	W	WATTS
CW	COLD WATER (POTABLE)	HST	HUMIDISTAT	PD	PRESSURE DROP	Wb	WET-BULB (TEMPERATURE)
CHWCC	CHILLED WATER COOLING COIL	HUM	HUMIDIFIER UNIT MOUNTED	PF	PRE-FILTER	WFCV	WATER FLOW CONTROL VALVE
		HVU	HEATING AND VENTILATING UNIT	PG	PRESSURE GAGE	WFM	WATER FLOW METER
		HW	HOT WATER	PHC	PREHEAT COIL PUMP	WFD	WATER FLOW MEASURING DEVICE
		HWC	HOT WATER COIL	PHD	PREHEAT COIL PUMP	WGD	WATER GAGE
		HWHU	HOT WATER UNIT HEATER	PPM	PARTS PER MILLION	XT	EXPANSION TANK
		HZ	HERTZ	PRV	PRESSURE REGULATING VALVE	YR	YEAR
		IO	INPUT/OUTPUT	PSI	POUNDS PER SQUARE INCH - ABSOLUTE		
		IAQ	INDOOR AIR QUALITY	PSIA	POUNDS PER SQUARE INCH - GAGE		
		IBT	INVERTED BUCKET TRAP	PSIG	POUNDS PER SQUARE INCH - GAGE		
		ICF	IN-LINE CENTRIFUGAL FAN	PSV	PRIMARY SECONDARY SYSTEM PRESSURE SAFETY VALVE		
		ICU	INTENSIVE CARE UNIT				
		ID	INSIDE DIAMETER				
		IFB	INTEGRAL FACE AND BYPASS				

MECHANICAL LEGEND



MECHANICAL GENERAL NOTES

- WORK SHALL CONFORM TO APPLICABLE LOCAL, STATE, AND NATIONAL CODES AND APPLICABLE VA STANDARDS.
- INSTALL EQUIPMENT IN ACCORDANCE WITH MANUFACTURER'S WRITTEN RECOMMENDATIONS.
- OBTAIN AND PAY FOR PERMITS, FEES, AND INSPECTIONS REQUIRED FOR THIS WORK.
- PROVIDE NEW MATERIALS, EQUIPMENT AND PRODUCTS INCORPORATED IN THE WORK, OF A SUITABLE GRADE FOR THE PURPOSES INTENDED.
- WHERE POSSIBLE, PROVIDE STANDARD PRODUCTS OF THE VARIOUS MANUFACTURERS EXCEPT WHERE SPECIAL CONSTRUCTION OR PERFORMANCE FEATURES ARE REQUIRED.
- EQUIPMENT OR MATERIAL DEVIATIONS FROM INDICATED CONTRACT REQUIREMENTS SHALL BE SUBJECT TO THE APPROVAL OF THE COR. SUBMIT PROPOSED EQUIPMENT DEVIATIONS THAT ARE SIMILAR BOTH IN QUALITY AND CAPACITY TO THAT EQUIPMENT SPECIFIED, AS SOLELY DETERMINED BY THE COR.
- PROVIDE MECHANICAL EQUIPMENT THAT IS LISTED AND LABELED BY UNDERWRITERS LABORATORIES (U.L.).
- INSTALL EQUIPMENT AS INDICATED, ALLOWING FOR SUFFICIENT ACCESS AND CLEARANCE SPACE FOR EQUIPMENT MAINTENANCE, REPAIRS, AND REPLACEMENT.
- INSTALL EQUIPMENT WITH PROPER CLEARANCES FOR REQUIRED PIPING AND ELECTRICAL SERVICES AND CONNECTIONS. INSTALL EQUIPMENT WITH REQUIRED ACCESS AND CLEARANCES IN ACCORDANCE WITH MANUFACTURER'S WRITTEN RECOMMENDATIONS AND APPLICABLE CODES AND STANDARDS.
- COORDINATE LOCATION, INSTALLATION, AND ROUTING OF DUCTWORK, PIPING, AND EQUIPMENT WITH WORK OF ALL OTHER TRADES.
- PROVIDE SUPPORT FOR MECHANICAL EQUIPMENT. COORDINATE SUPPORT LOCATIONS WITH ALL CONTRACTORS PRIOR TO INSTALLATION.
- PROVIDE POWER CONNECTIONS TO EQUIPMENT PROVIDED UNDER THIS CONTRACT AS WORK OF DIVISION 26.
- PROVIDE CONTROL WIRING FOR MECHANICAL EQUIPMENT AS WORK OF DIVISION 23.
- DUCTWORK AND PIPING LAYOUTS AND LOCATIONS ARE SCHEMATIC. DO NOT SCALE THESE DRAWINGS. COORDINATE AND DETERMINE EXACT ROUTING OF DUCTWORK AND PIPING IN THE FIELD.
- FIELD VERIFY DIMENSIONS BY ACTUAL MEASUREMENT AND OBSERVATION BEFORE ORDERING OR FABRICATING DUCTWORK, PIPING OR EQUIPMENT.
- NOTIFY COR REGARDING DISCREPANCIES BETWEEN CONTRACT DOCUMENTS AND EXISTING CONDITIONS OR DIMENSIONS BEFORE PERFORMING WORK. FAILURE TO VERIFY AND REPORT SHALL CONSTITUTE CONTRACTOR'S ACCEPTANCE OF EXISTING CONDITIONS AS FIT FOR THE PROPER EXECUTION OF THE WORK.
- SEE ARCHITECTURAL DRAWINGS FOR FINAL LOCATION OF CEILING INSTALLED EQUIPMENT.
- INSTALL DUCTWORK AND PIPING AS HIGH AS POSSIBLE AND AS CLOSE AS POSSIBLE TO BUILDING WALLS, CEILING AND FLOOR AND ROOF STRUCTURE. PROVIDE OFFSETS, FITTINGS, AND DIMENSIONAL TRANSITIONS NOT SHOWN BUT REQUIRED TO MAINTAIN MINIMUM CLEAR SPACE TO BUILDING SURFACES AT NO ADDITIONAL COST.
- PATCH, PAINT AND CLEAN SURFACES SUSTAINING DAMAGE ASSOCIATED WITH MECHANICAL WORK OF THIS CONTRACT, UNLESS NOTED OTHERWISE.
- SUBMIT BOUND SET OF OPERATING AND MAINTENANCE INSTRUCTIONS FOR EQUIPMENT AT COMPLETION OF PROJECT.
- INSTALL ESCUTCHEONS IN ALL PLACES WHERE PIPING PENETRATES A WALL IN AN EXPOSED LOCATION.
- MAKE A COMPLETE REVIEW OF DRAWINGS, INCLUDING SCHEDULES AND DETAILS, PRIOR TO BEGINNING MECHANICAL SYSTEM INSTALLATION. RESOLVE CONFLICTS WITH COR.
- DRAWING PLANS, SCHEMATICS, AND DIAGRAMS INDICATE GENERAL LOCATION AND ARRANGEMENT OF PIPING SYSTEMS.
- INSTALL PIPING INDICATED TO BE EXPOSED AND PIPING IN EQUIPMENT ROOMS AND SERVICE AREAS AT RIGHT ANGLES OR PARALLEL TO BUILDING WALLS. DIAGONAL RUNS ARE NOT ACCEPTABLE UNLESS SPECIFICALLY INDICATED OTHERWISE.
- INSTALL PIPING FREE OF SAGS AND BENDS.
- INSTALL PIPING TO ALLOW APPLICATION OF INSULATION.
- PIPING SHALL NOT BE ROUTED OVER TELECOM OR ELECTRICAL CLOSETS/ROOMS.
- PREPARE DRAWINGS SHOWING MECHANICAL SYSTEMS AND EQUIPMENT FOR COORDINATION WITH WORK OF OTHER DIVISIONS AND WORK BY THE GOVERNMENT.
 - INCLUDE PLAN VIEWS, SECTIONS, AND ELEVATIONS, AS REQUIRED, TO FULLY COORDINATE ALL NEW WORK WITH ITSELF AND EXISTING CONDITIONS.
 - DRAWINGS SHALL SHOW, BUT NOT BE LIMITED TO, ALL DUCTWORK, AIR DISTRIBUTION, MECHANICAL EQUIPMENT, MECHANICAL PIPING, FIRE PROTECTION PIPING, PLUMBING PIPING, CABLE TRAYS, LIGHTING FIXTURES, CEILING GRID AND HEIGHT, BEAMS AND JOISTS (WITH ELEVATIONS MARKED), ELECTRICAL CONDUIT LARGER THAN 2 INCHES IN DIAMETER AND OTHER CEILING MOUNT DEVICES OR EQUIPMENT THAT PROTRUDE INTO THE CEILING CAVITY.
 - IF THERE ARE OUTSTANDING ISSUES THAT CANNOT BE RESOLVED, CONSULT WITH COR FOR GUIDANCE AND MAKE CORRECTIONS IN ACCORDANCE WITH DIRECTIONS GIVEN.
 - DO NOT BEGIN FABRICATION OR PLACE EQUIPMENT ORDERS UNTIL COORDINATION DRAWINGS HAVE BEEN APPROVED.
 - INSTALLATION COMMENCED PRIOR TO COORDINATION DRAWING APPROVAL IS TAKEN AT THE CONTRACTORS OWN RISK. MODIFY, MOVE OR RECONFIGURE WORK LACKING COORDINATION AT NO ADDITIONAL COST.

HVAC DESIGN DATA

Key Name	SUMMER			WINTER			LOWEST AVERAGE ANNUAL DEWPOINT
	DRY BULB	WET BULB	% HUMIDITY	TEMP	DEWPOINT	% HUMIDITY	
1	97.4 °F	77.8 °F	42.1	15.9 °F	9.0 °F	10	3
2	75.0 °F	65.3 °F	60	70.0 °F	50.0 °F	20	28
3	75.0 °F	65.3 °F	60	70.0 °F	50.0 °F	20	28
4	75.0 °F	65.3 °F	60	70.0 °F	50.0 °F	20	28
5	75.0 °F	65.3 °F	60	70.0 °F	50.0 °F	20	28
6	75.0 °F	65.3 °F	60	70.0 °F	50.0 °F	20	28
7	75.0 °F	65.3 °F	60	70.0 °F	50.0 °F	20	28
8	75.0 °F	65.3 °F	60	70.0 °F	50.0 °F	20	28
10	75.0 °F	65.3 °F	60	70.0 °F	50.0 °F	20	28
12	75.0 °F	65.3 °F	60	70.0 °F	50.0 °F	20	28
13	75.0 °F	65.3 °F	60	70.0 °F	50.0 °F	20	28
14	75.0 °F	65.3 °F	60	70.0 °F	50.0 °F	20	28
15	75.0 °F	65.3 °F	60	70.0 °F	50.0 °F	20	28
16	75.0 °F	65.3 °F	60	70.0 °F	50.0 °F	20	28
17	75.0 °F	65.3 °F	60	70.0 °F	50.0 °F	20	28
18	75.0 °F	65.3 °F	60	70.0 °F	50.0 °F	20	28
19	75.0 °F	65.3 °F	60	70.0 °F	50.0 °F	20	28
20	75.0 °F	65.3 °F	60	70.0 °F	50.0 °F	20	28

- NOTES:
- ELECTRICAL EQUIPMENT ROOMS WITHOUT INTERNAL HEAT GAIN HAVE NO TEMPERATURE REQUIREMENTS. ELECTRICAL EQUIPMENT ROOMS WITH HEAT GAIN ARE REQUIRED TO BE 86 SUMMER/40 WINTER, BUT HAVE BEEN DESIGNED TO ADJACENT SPACE TEMPERATURES DUE TO THEIR SMALL SIZE.
 - SOILED ROOMS HAVE NO TEMPERATURE REQUIREMENTS, BUT HAVE BEEN DESIGNED TO THE TEMPERATURES INDICATED.

FULLY SPRINKLERED
100% CONSTRUCTION DOCUMENTS

Revisions:

Date

CONSULTANTS:

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PROJECT MANAGER

ACG Project Number
16-352

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Office of
Construction
and Facilities
Management

U.S. Department of
Veterans Affairs

Drawing Title
**MECHANICAL LEGEND,
ABBREVIATIONS, AND GENERAL
NOTES**

Location
**2002 HOLCOMBE BOULEVARD,
HOUSTON, TX 77030 4298**

Project Title
**RENOVATE/EXPAND
EMERGENCY ROOM SYSTEM**

Approved: Mark Landwert - COR

Date
MARCH 15, 2019

Checked
SPG

Drawn
WRS

VA PROJECT NUMBER
580-17-101

Building Number
1

Drawing Number
M-001

Dwg 1 of 12

A

B

C

D

E

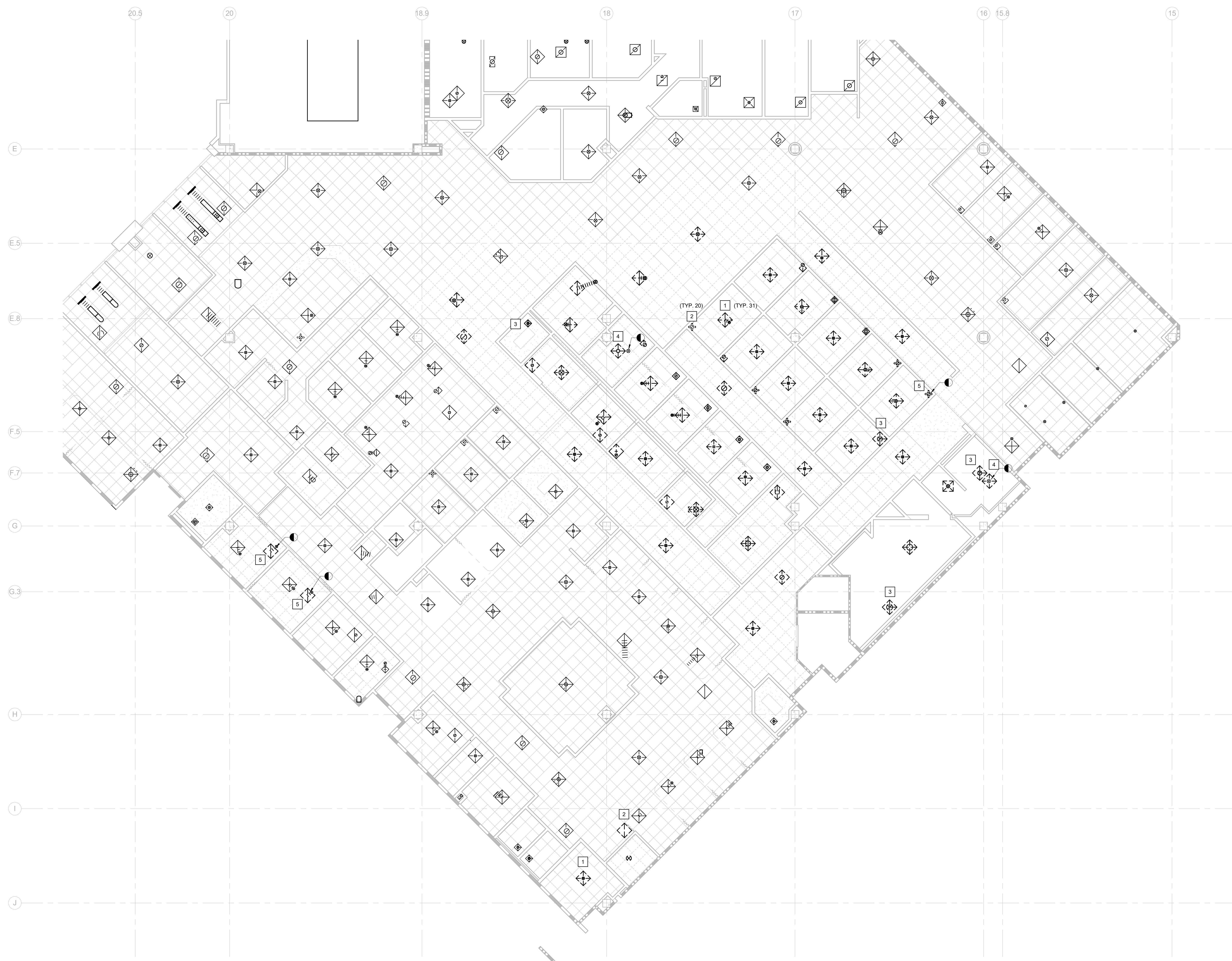
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B

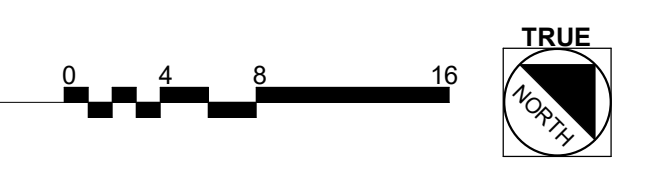
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E



1 FIRST FLOOR CEILING HVAC DEMOLITION PLAN
SCALE: 1/8" = 1'-0"



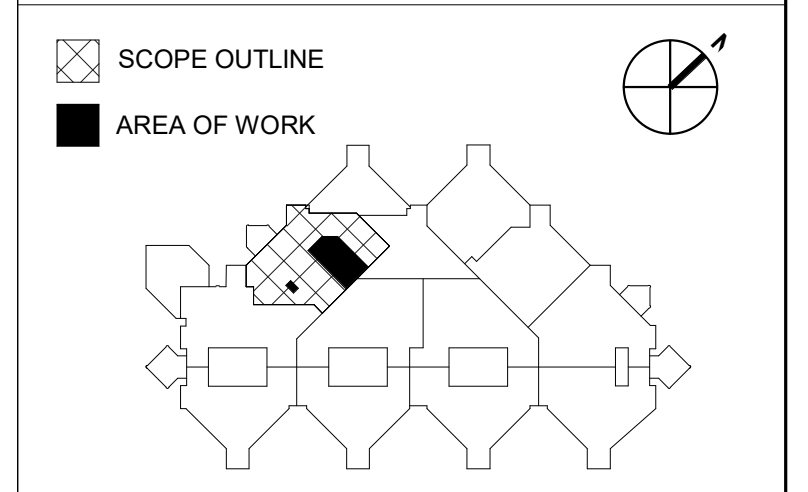
SHEET NOTES:

- COORDINATE ALL WORK WITH OTHER TRADES AND THE COR TO MINIMIZE DISRUPTION OF AREAS OUTSIDE OF THE PROJECT LIMITS OF WORK.
- ALL DEMOLISHED EQUIPMENT SHALL BECOME THE PROPERTY OF THE CONTRACTOR UNLESS NOTED OTHERWISE. CONTRACTOR SHALL DISPOSE OF EQUIPMENT IN ACCORDANCE WITH ALL VAMC REQUIREMENTS AND ENVIRONMENTAL REGULATIONS.
- CONTRACTOR TO FIELD VERIFY ALL EXISTING CONDITIONS. PRESERVE ANY EXISTING DUCTWORK, PIPING OR EQUIPMENT THAT IS NOT INDICATED FOR DEMOLITION OR IS DISCOVERED IN THE DEMOLITION PROCESS.
- CAP EXISTING DUCTWORK AND PIPING AT THE LIMITS OF DEMOLITION.
- UNLESS OTHERWISE NOTED, WORK IS TO BE COMPLETED AS A PART OF PHASE 1. REFER TO SHEET G103 FOR PHASING AREA DETAILS.
- REFER TO SHEET G104 FOR DEDUCT ALTERNATE DETAILS.

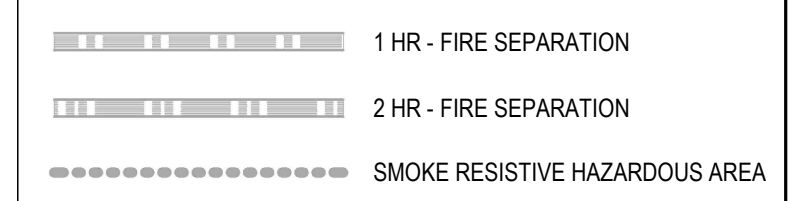
SHEET KEY NOTES:

- DEMOLISH SUPPLY AIR DIFFUSER AND ASSOCIATED DUCTWORK TO LIMITS OF DEMOLITION ON FIRST FLOOR INTERSTITIAL. SEE MD102.
- DEMOLISH RETURN AIR GRILLE AND ASSOCIATED DUCTWORK TO LIMITS OF DEMOLITION ON FIRST FLOOR INTERSTITIAL. SEE MD102.
- DEMOLISH EXHAUST GRILL AND ASSOCIATED DUCTWORK TO LIMITS OF DEMOLITION.
- DEMOLISH SUPPLY AIR DIFFUSER AND ASSOCIATED DUCTWORK TO LIMITS OF DEMOLITION.
- DEMOLISH RETURN AIR GRILLE AND ASSOCIATED DUCTWORK TO LIMITS OF DEMOLITION.

KEY PLAN



WALL RATING LEGEND

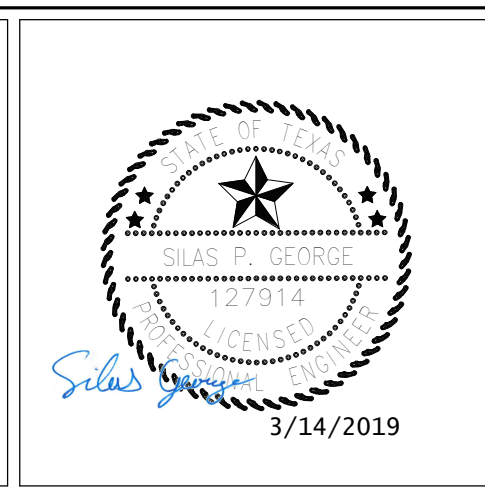


FULLY SPRINKLERED
100% CONSTRUCTION DOCUMENTS

Revisions:	Date

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VA

Drawing Title
**FIRST FLOOR CEILING HVAC
DEMOLITION PLAN**

Location
**2002 HOLCOMBE BOULEVARD,
HOUSTON, TX 77030 4298**

Project Title
**RENOVATE/EXPAND
EMERGENCY ROOM SYSTEM**

Approved: Mark Landwert - COR

Date MARCH 15, 2019

Checked SPG

Drawn WRS

VA PROJECT NUMBER
580-17-101

Building Number
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MD101

Dwg 2 of 12

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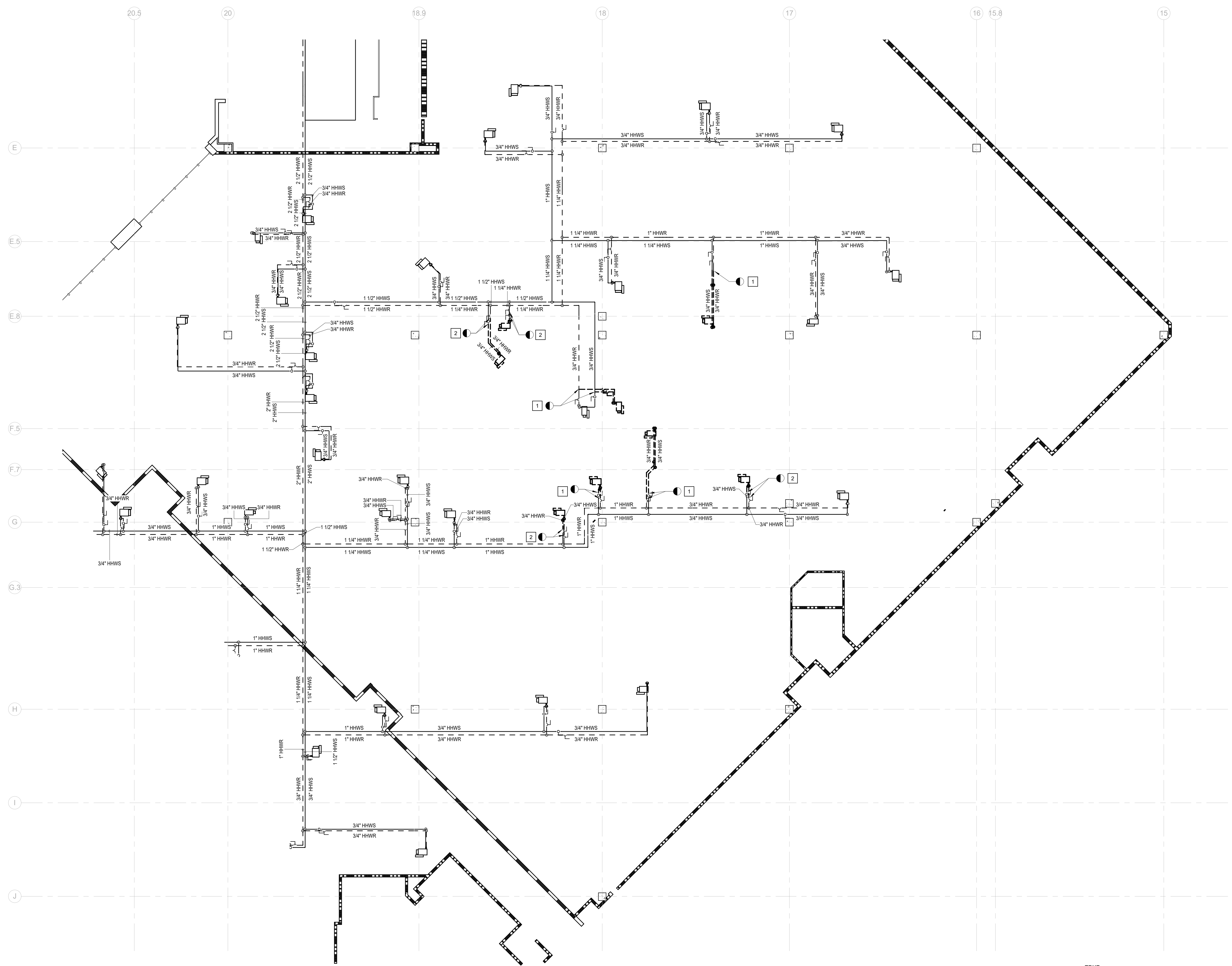
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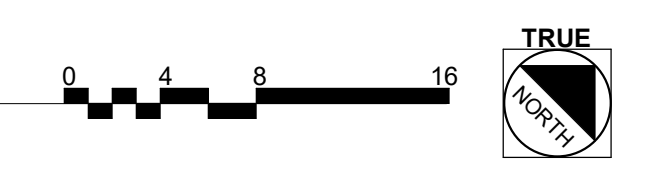
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E



1 FIRST FLOOR INTERSTITIAL MECHANICAL PIPING DEMOLITION PLAN
SCALE: 1/8" = 1'-0"



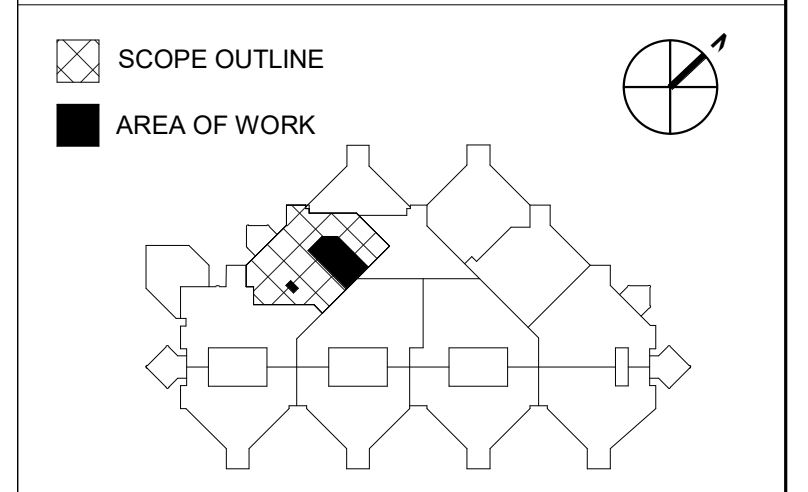
SHEET NOTES:

- COORDINATE ALL WORK WITH OTHER TRADES AND THE COR TO MINIMIZE DISRUPTION OF AREAS OUTSIDE OF THE PROJECT LIMITS OF WORK.
- ALL DEMOLISHED EQUIPMENT SHALL BECOME THE PROPERTY OF THE CONTRACTOR UNLESS NOTED OTHERWISE. CONTRACTOR SHALL DISPOSE OF EQUIPMENT IN ACCORDANCE WITH ALL VMC REQUIREMENTS AND ENVIRONMENTAL REGULATIONS.
- CONTRACTOR TO FIELD VERIFY ALL EXISTING CONDITIONS. PRESERVE ANY EXISTING DUCTWORK, PIPING OR EQUIPMENT THAT IS NOT INDICATED FOR DEMOLITION OR IS DISCOVERED IN THE DEMOLITION PROCESS.
- CAP EXISTING DUCTWORK AND PIPING AT THE LIMITS OF DEMOLITION.
- UNLESS OTHERWISE NOTED, WORK IS TO BE COMPLETED AS A PART OF PHASE 1. REFER TO SHEET G1103 FOR PHASING AREA DETAILS.
- REFER TO SHEET G1104 FOR DEDUCT ALTERNATE DETAILS.

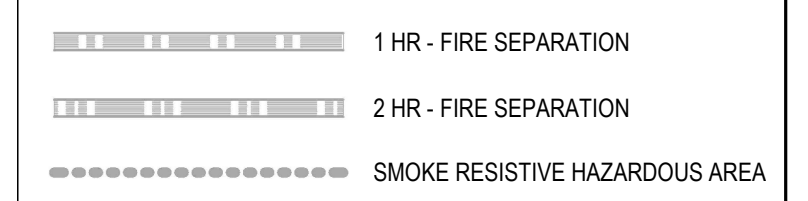
SHEET KEY NOTES:

- DEMOLISH VAV. AND HEATING HOT WATER PIPING TO LIMITS OF DEMOLITION.
- DEMOLISH VAV AND HEATING HOT WATER PIPING TO LIMITS OF DEMOLITION. CAP PIPES AT VALVES.

KEY PLAN

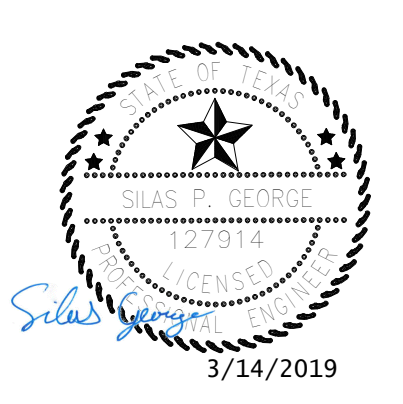


WALL RATING LEGEND



FULLY SPRINKLERED
100% CONSTRUCTION DOCUMENTS

CONSULTANTS:



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Office of
Construction
and Facilities
Management



Drawing Title

FIRST FLOOR INTERSTITIAL
MECHANICAL PIPING DEMOLITION
PLAN

Location

2002 HOLCOMBE BOULEVARD,
HOUSTON, TX 77030 4298

Project Title

RENOVATE/EXPAND
EMERGENCY ROOM SYSTEM

Approved: Mark Landwert - COR

Date
MARCH 15, 2019

Checked
SPG

Drawn
WRS

VA PROJECT NUMBER
580-17-101

Building Number
1

Drawing Number
MD103

Dwg 4 of 12

A

B

C

D

E

A

B

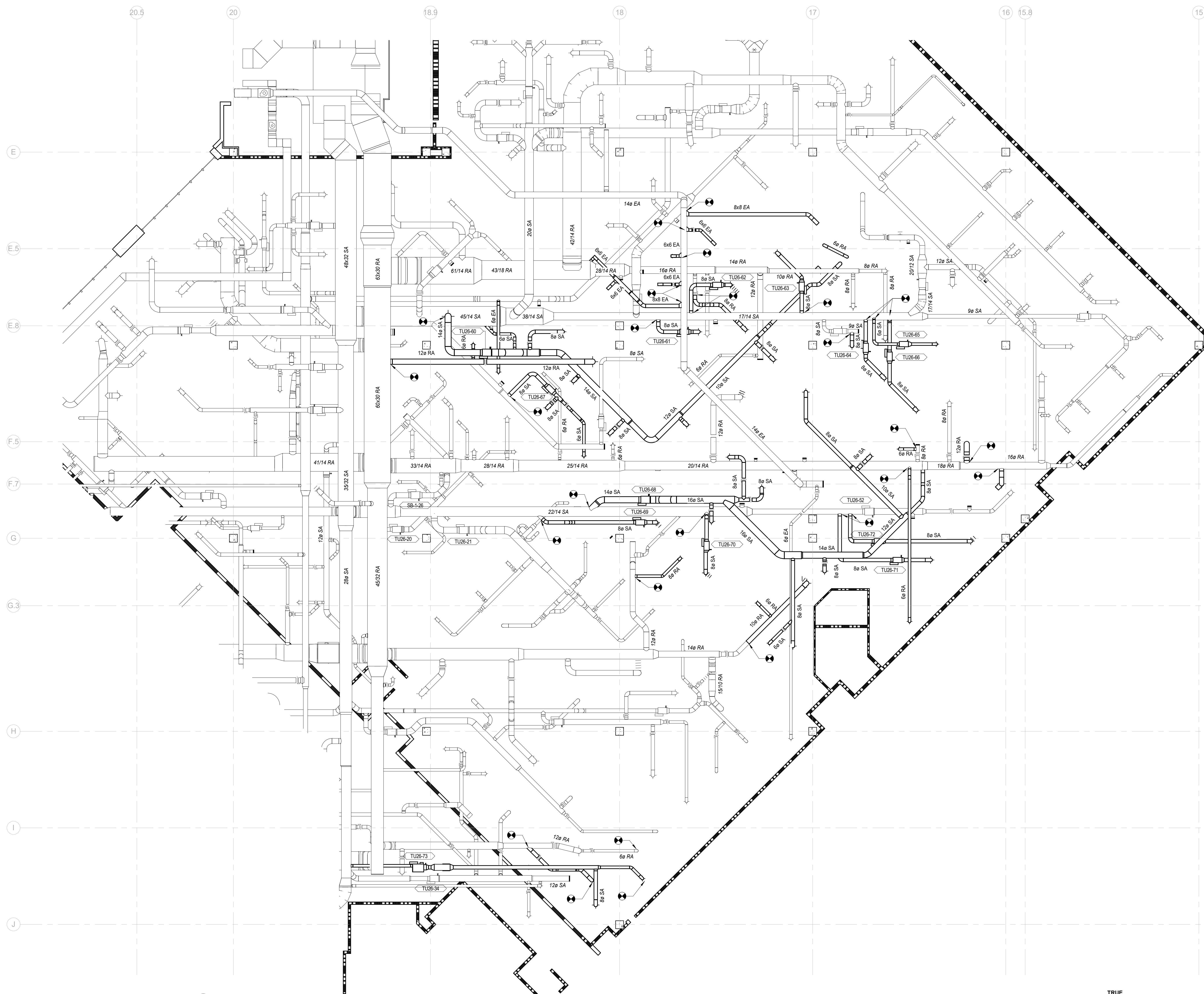
C

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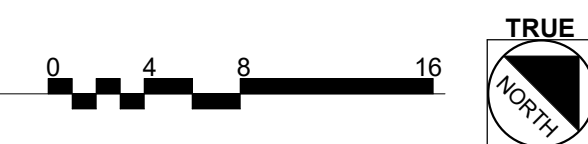
E

SHEET NOTES:

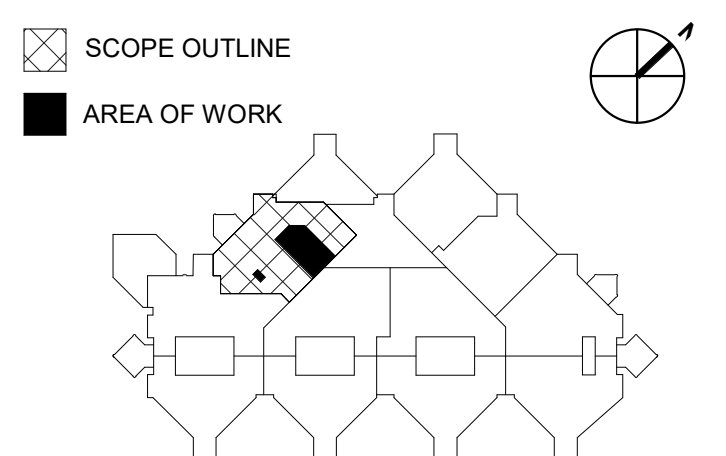
- A COORDINATE ALL WORK WITH OTHER TRADES AND THE COR TO MINIMIZE DISRUPTION OF AREAS OUTSIDE OF THE PROJECT LIMITS OF WORK.
- B UNLESS NOTED OTHERWISE, ALL DIFFUSERS AND GRILLE RUN-OUTS ARE THE SAME SIZE AS ITS NECK/DUCT CONNECTION SPECIFIED IN THE AIR DEVICE SCHEDULE.
- C ALL DIFFUSER AND GRILLE TAKEOFFS SHALL INCLUDE A BALANCING DAMPER.
- D SEE ARCHITECTURAL CEILING PLAN FOR CEILING TYPES AS WELL AS DIFFUSER AND GRILLE PLACEMENT WITHIN THE CEILING.
- E REUSE EXISTING INTERSTITIAL FLOOR PENETRATIONS.



1 FIRST FLOOR INTERSTITIAL COMPOSITE HVAC PLAN
SCALE: 1/8" = 1'-0"



KEY PLAN

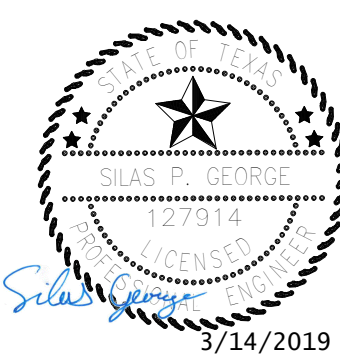


WALL RATING LEGEND

- 1 HR - FIRE SEPARATION
- 2 HR - FIRE SEPARATION
- SMOKE RESISTIVE HAZARDOUS AREA

FULLY SPRINKLERED
100% CONSTRUCTION DOCUMENTS

CONSULTANTS:



PROJECT MANAGER

Headquarters
Apogee Consulting Group, P.A.
1151 Kildaire Farm Road, Suite 120
Cary, North Carolina 27511
www.acg-pa.com 919-858-7420

Office of
Construction
and Facilities
Management



Drawing Title
**FIRST FLOOR INTERSTITIAL
COMPOSITE HVAC PLAN**

Location
**2002 HOLCOMBE BOULEVARD,
HOUSTON, TX 77030 4298**

Project Title
**RENOVATE/EXPAND
EMERGENCY ROOM SYSTEM**

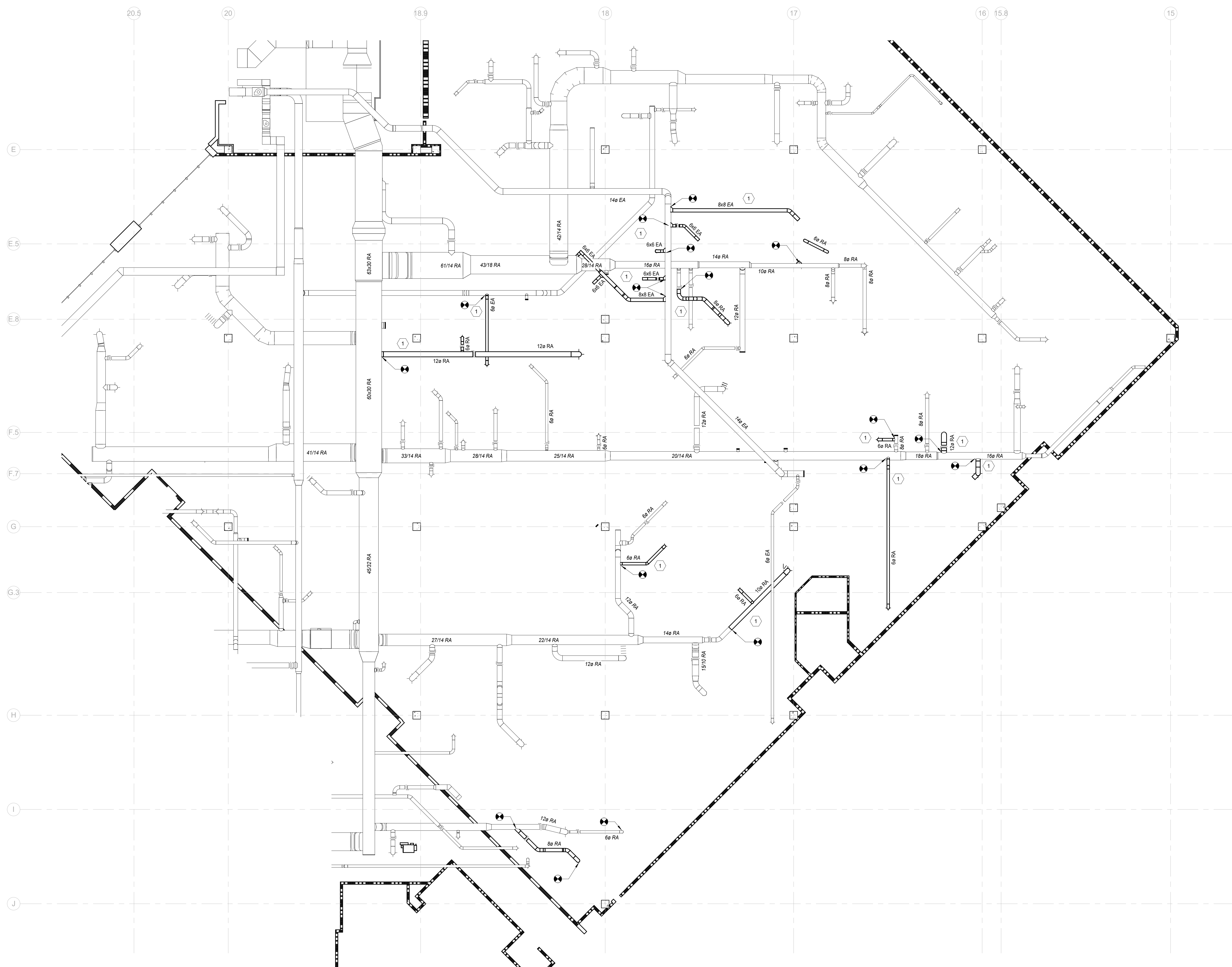
Approved: Mark Landwert - COR

Date
MARCH 15, 2019

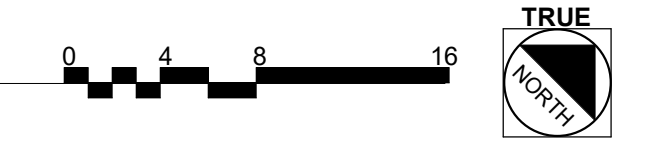
Checked
SPG

Drawn
WRS

VA PROJECT NUMBER
580-17-101
Building Number
1
Drawing Number
MH102
Dwg 6 of 12



1 FIRST FLOOR INTERSTITIAL HVAC EXHAUST/RETURN PLAN
SCALE: 1/8" = 1'-0"



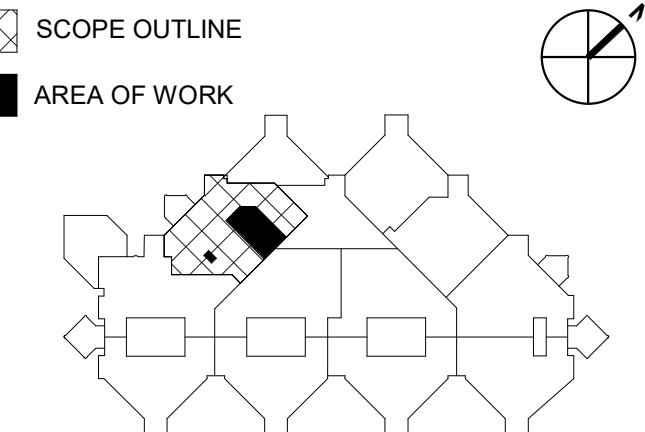
SHEET NOTES:

- A COORDINATE ALL WORK WITH OTHER TRADES AND THE COR TO MINIMIZE DISRUPTION OF AREAS OUTSIDE OF THE PROJECT LIMITS OF WORK.
- B UNLESS NOTED OTHERWISE, ALL DIFFUSERS AND GRILLE RUN-OUTS ARE THE SAME SIZE AS ITS NECK/DUCT CONNECTION SPECIFIED IN THE AIR DEVICE SCHEDULE.
- C ALL DIFFUSER AND GRILLE TAKEOFFS SHALL INCLUDE A BALANCING DAMPER.
- D SEE ARCHITECTURAL CEILING PLAN FOR CEILING TYPES AS WELL AS DIFFUSER AND GRILLE PLACEMENT WITHIN THE CEILING.
- E REUSE EXISTING INTERSTITIAL FLOOR PENETRATIONS.

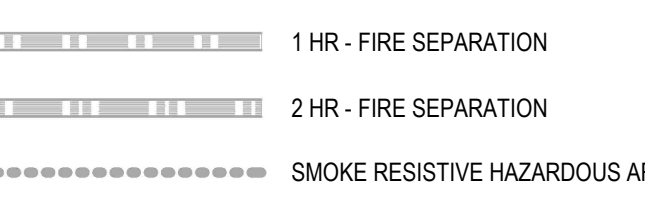
SHEET KEY NOTES:

- 1 CONNECT NEW DUCT TO EXISTING WHERE INDICATED. REFER TO MH102 FOR SUPPLY DUCTS TO COORDINATE ROUTING.

KEY PLAN



WALL RATING LEGEND



FULLY SPRINKLERED
100% CONSTRUCTION DOCUMENTS

CONSULTANTS:

rpadesign
5960 Fairview Road, Suite 500
Charlotte, NC 28210
704-887-3500



PROJECT MANAGER

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1151 Kildaire Farm Road, Suite 120
Cary, North Carolina 27511

ACG Project Number
16-352

Engineers | Architects www.acg-pa.com 919-858-7420

Office of
Construction
and Facilities
Management

VA U.S. Department of Veterans Affairs

Drawing Title
FIRST FLOOR INTERSTITIAL HVAC
EXHAUST/RETURN PLAN

Location
2002 HOLCOMBE BOULEVARD,
HOUSTON, TX 77030 4298

Project Title
RENOVATE/EXPAND
EMERGENCY ROOM SYSTEM

Approved: Mark Landwert - COR

Date
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Checked
SPG

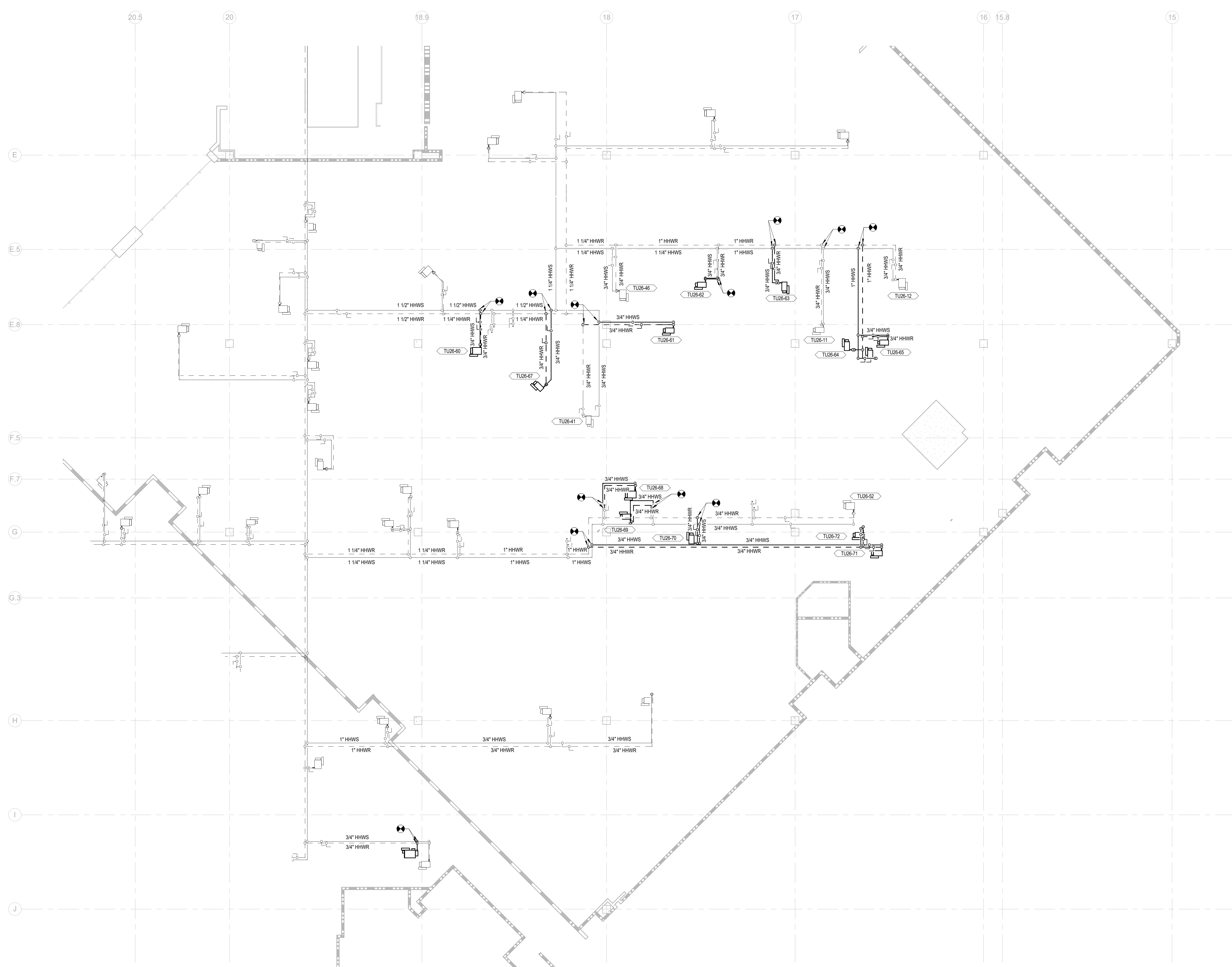
Drawn
WRS

VA PROJECT NUMBER
580-17-101

Building Number
1

Drawing Number
MH104

Dwg 8 of 12



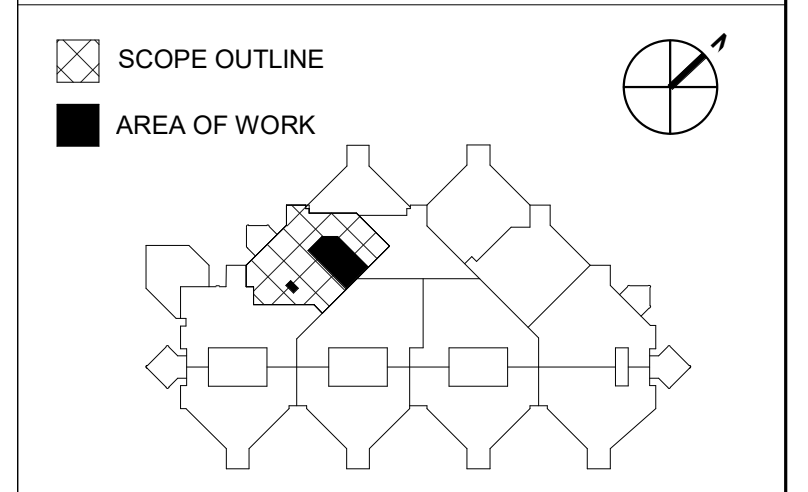
1 FIRST FLOOR INTERSTITIAL MECHANICAL PIPING PLAN
SCALE: 1/8" = 1'-0"



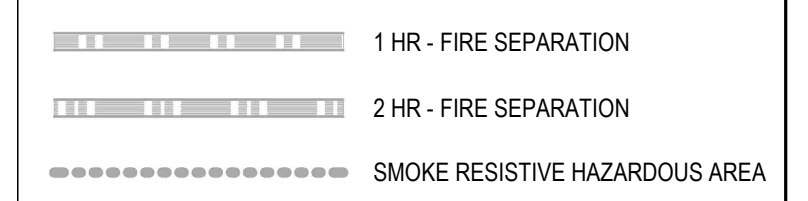
SHEET NOTES:

- 1 COORDINATE ALL WORK WITH THE GENERAL CONTRACTOR, OTHER TRADES AND THE COR TO MINIMIZE DISRUPTION OF AREAS OUTSIDE OF THE PROJECT LIMITS OF WORK.
- 2 UNLESS OTHERWISE NOTED, WORK IS TO BE COMPLETED AS A PART OF PHASE 1. REFER TO SHEET G103 FOR PHASING AREA DETAILS.

KEY PLAN



WALL RATING LEGEND



**FULLY SPRINKLERED
100% CONSTRUCTION DOCUMENTS**

CONSULTANTS:



PROJECT MANAGER

Headquarters
Apogee Consulting Group, P.A.
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www.acg-pa.com 919-858-7420

ACG Project Number
16-352

Office of
Construction
and Facilities
Management



Drawing Title
**FIRST FLOOR INTERSTITIAL
MECHANICAL PIPING PLAN**

Location
**2002 HOLCOMBE BOULEVARD,
HOUSTON, TX 77030 4298**

Project Title
**RENOVATE/EXPAND
EMERGENCY ROOM SYSTEM**

Approved: Mark Landwert - COR

Date
MARCH 15, 2019

Checked
SPG

Drawn
WRS

VA PROJECT NUMBER
580-17-101

Building Number
1

Drawing Number
MP101

Dwg 9 of 12

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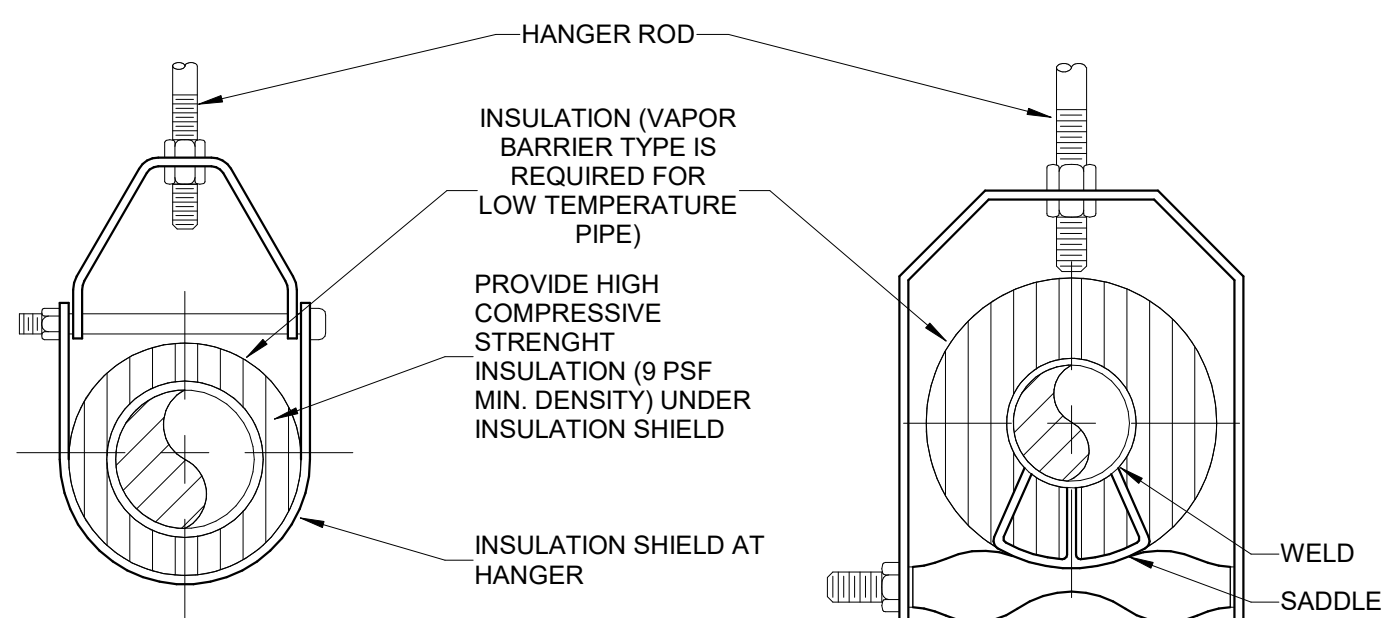
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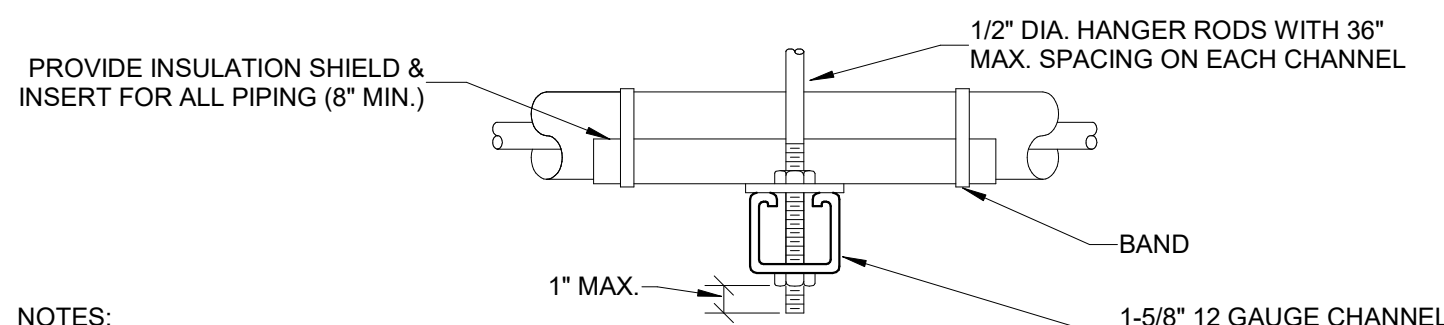
D

E



ADJUSTABLE CLEVIS HANGER TYPE 1 - SEE SPECIFICATIONS

ADJUSTABLE CLEVIS HANGER TYPE 43 - SEE SPECIFICATIONS



NOTES:
SEE SPECIFICATIONS FOR
DETAILED HANGER
REQUIREMENTS.

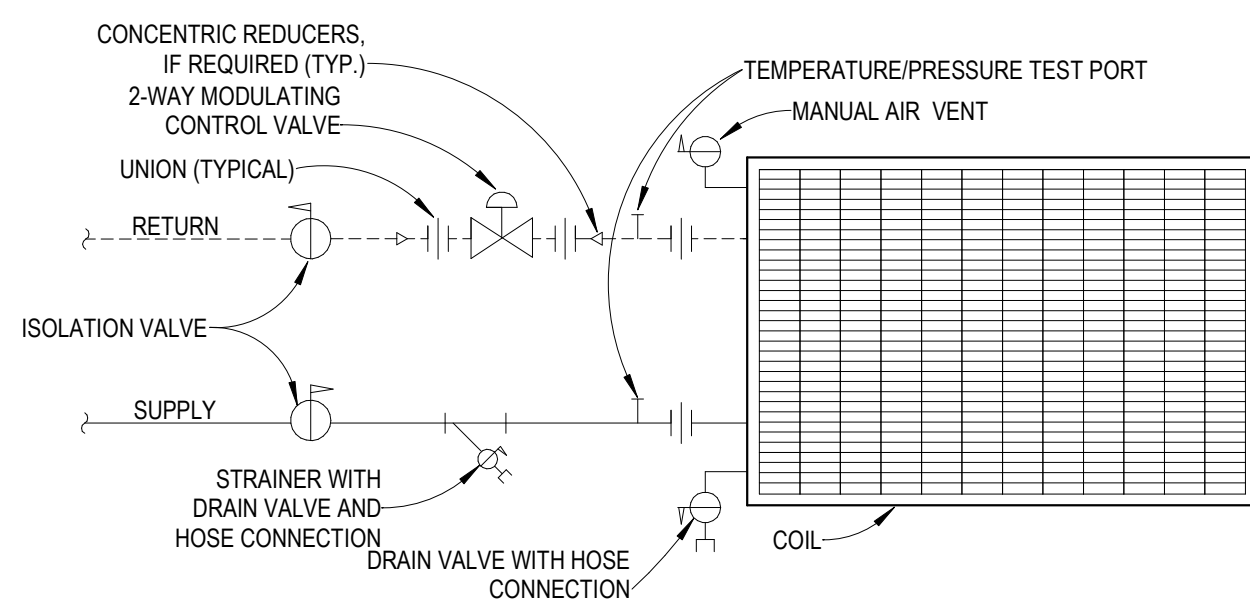
SIDE VIEW TRAPEZE HANGER
FOR UP TO 1000 LB. UNIFORM
LOAD.

MAXIMUM PIPE/TUBING SUPPORT SPACING													
NOM. SIZE (IN.)	THRU 3/4	1	1 1/4	1 1/2	2	2 1/2	3	4	5	6	8	10	12
PIPE (FT.)	7	7	7	6	10	11	12	14	16	17	19	22	23
TUBING (FT.)	5 FT	6	7	6	6	6	10	12	13	14	16	-	-

NOTE: FOR TRAPEZE HANGER TAKE SPACINGS OF SMALLEST SIZE ON TRAPEZE

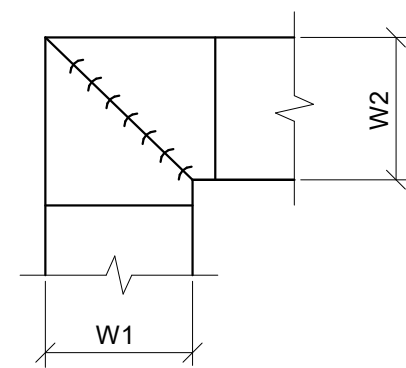
7 PIPE HANGER

SCALE: N.T.S.



8 VAV COIL PIPING DETAIL

SCALE: N.T.S.

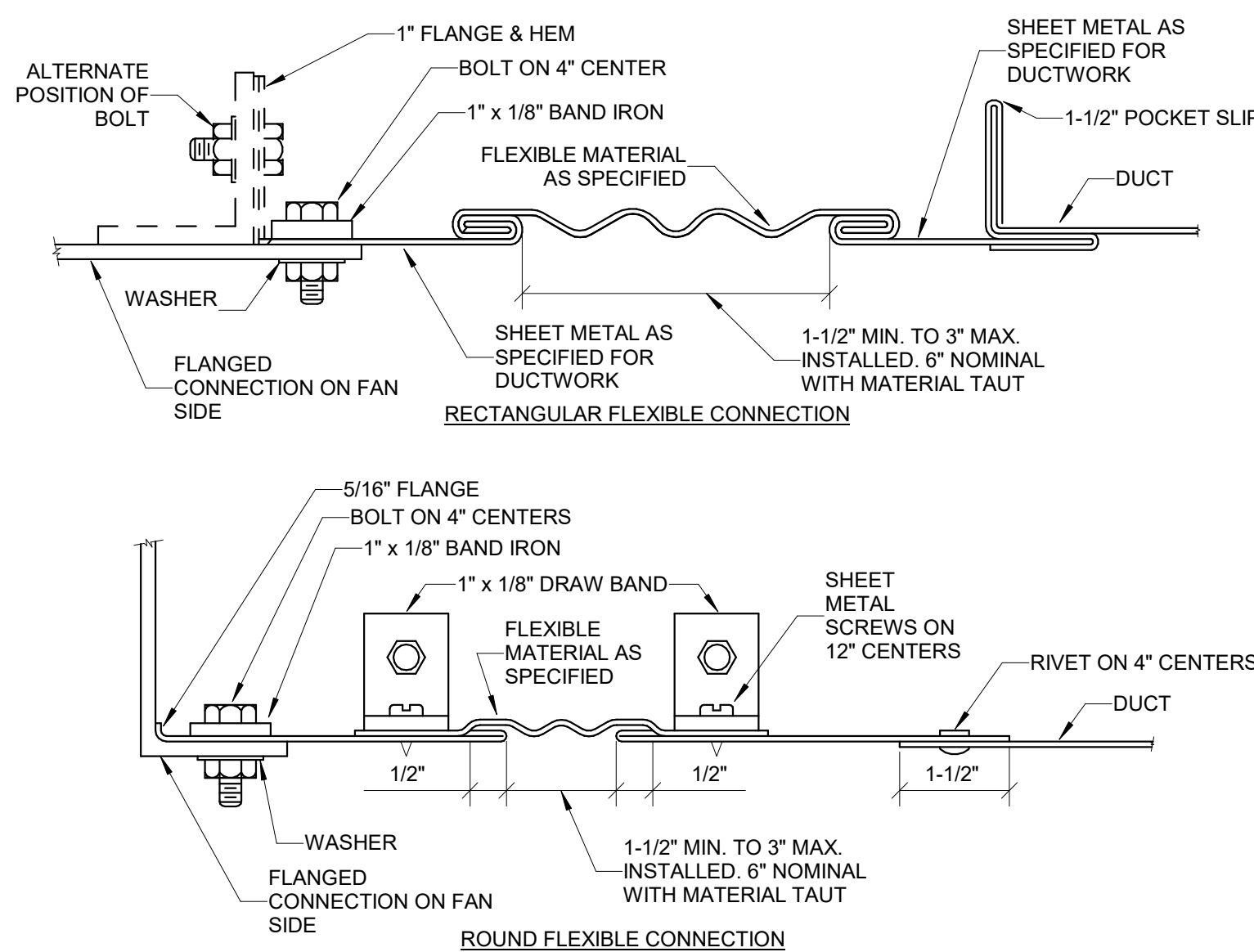


NOTES:

- ALL VANE ELBOWS SHALL BE CONSTRUCTED AND INSTALLED AS DETAILED BY SMACNA.
- WHEN W1 DOES NOT EQUAL W2, VANE SHALL BE SINGLE THICKNESS VANE TYPE REGARDLESS OF W DIMENSION.
- ALL SINGLE THICKNESS VANES SHALL HAVE A 2" RADIUS, 1-1/2" MAXIMUM SPACE BETWEEN VANES AND A 3/4" TRAILING EDGE.
- WHEN W EQUAL W2 AND W1 IS GREATER THAN 20" VANES SHALL BE DOUBLE VANE TYPE.

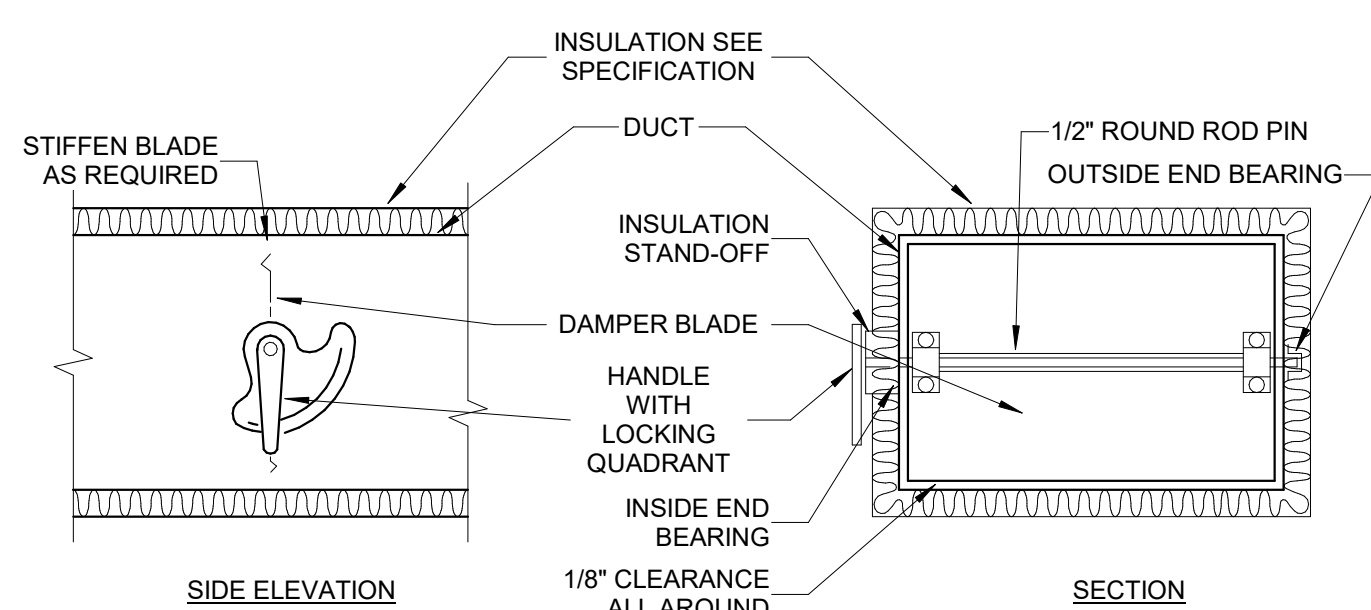
4 DUCTWORK SQUARE VANE ELBOWS

SCALE: N.T.S.



5 FLEXIBLE DUCT CONNECTIONS

SCALE: N.T.S.

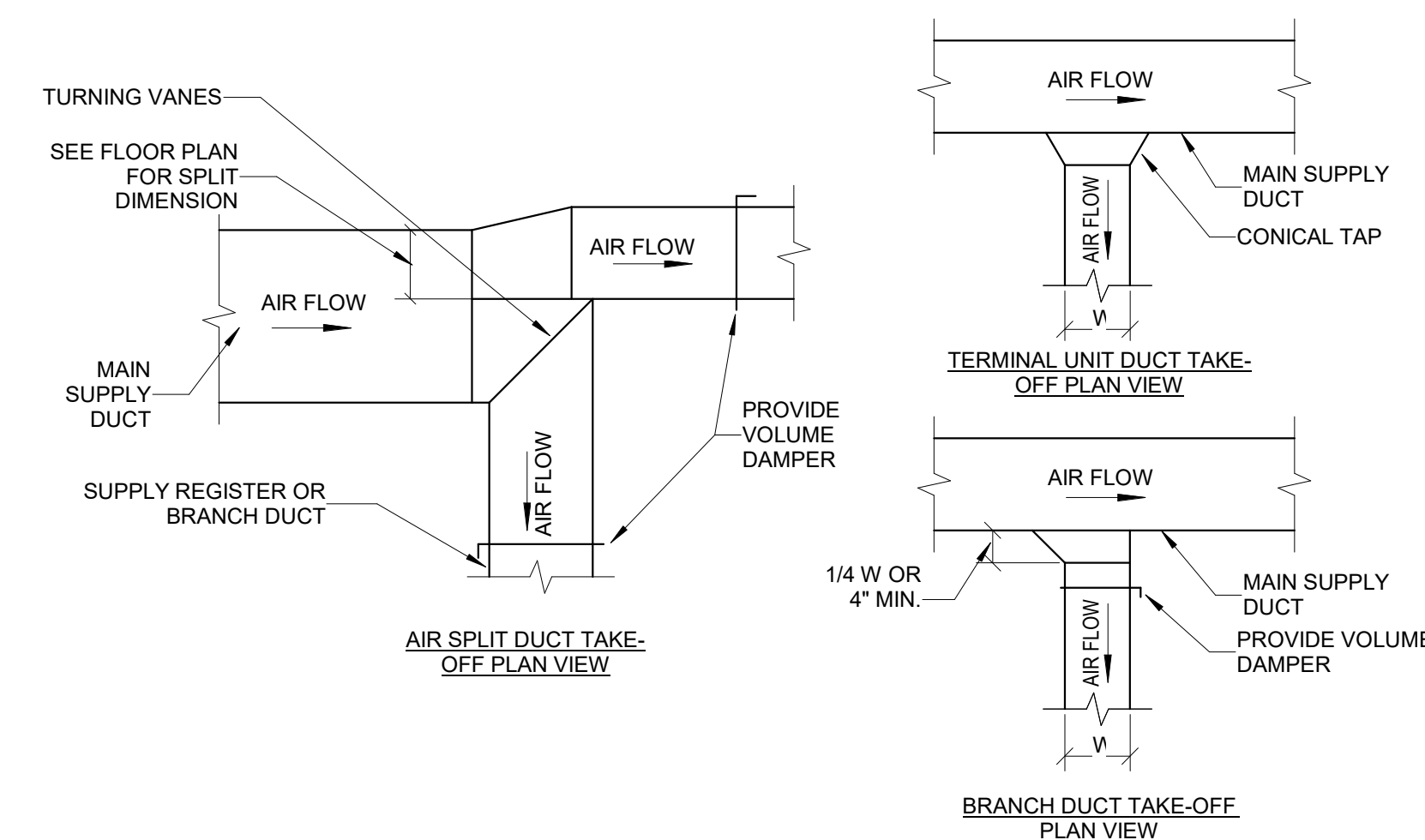


NOTES:

- DELETE INSULATION STAND-OFF ON DUCTWORK WITHOUT EXTERIOR INSULATION.
- DETAIL SHOWS SINGLE BLADE DAMPER, DAMPER INSTALLATION SHALL BE SIMILAR FOR MULTI-BLADE DAMPERS & ROUND DAMPERS.

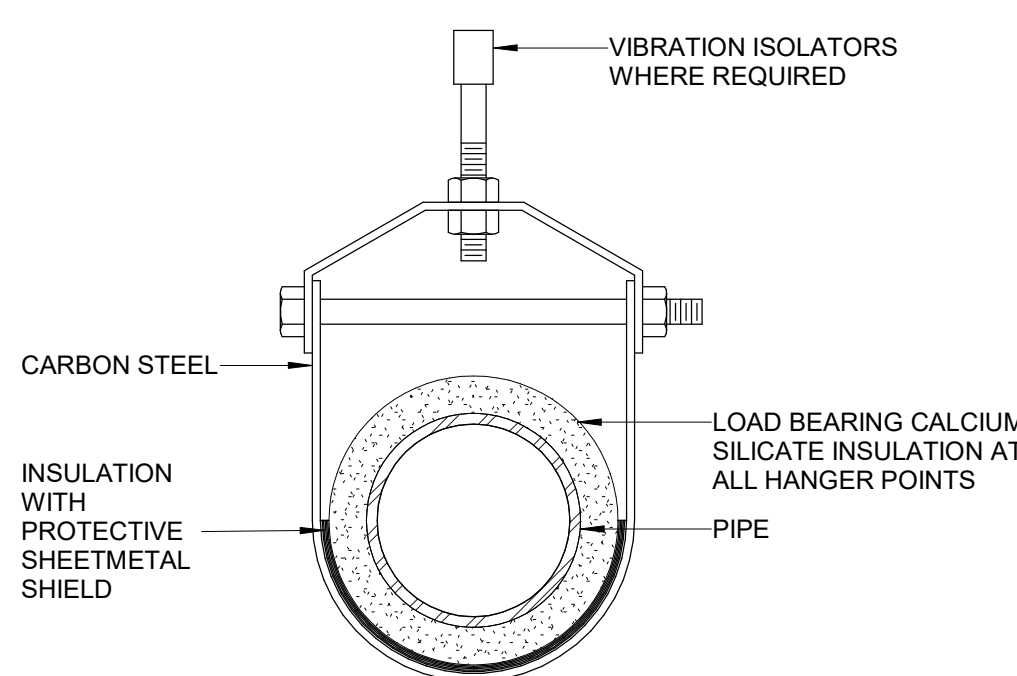
1 VOLUME DAMPER DETAIL

SCALE: N.T.S.



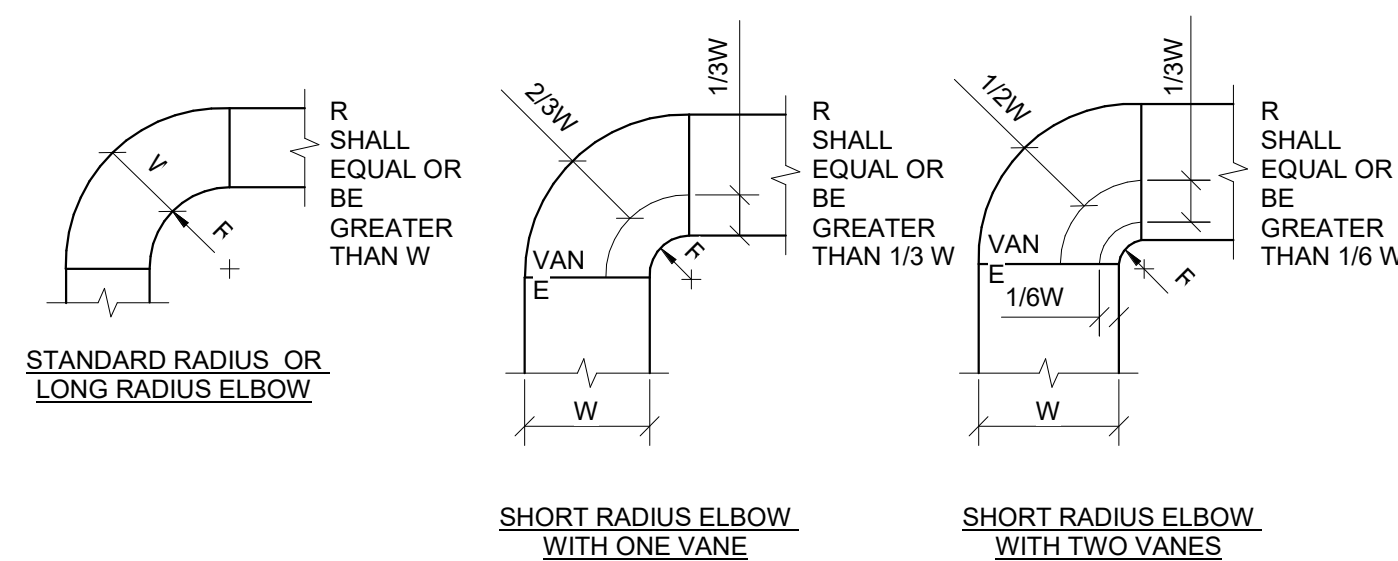
2 SUPPLY DUCT TAKE-OFFS

SCALE: N.T.S.



6 CLEVIS PIPE HANGER DETAIL

SCALE: N.T.S.



NOTES:

- THE INTERIOR SURFACE OF ALL RADIUS ELBOWS SHALL BE MADE ROUND.
- ALL STANDARD ELBOWS CAN BE SUBSTITUTED WITH SHORT RADIUS ELBOWS. ALL SHORT RADIUS ELBOWS SHALL HAVE VANES. VANES SHALL BE CONSTRUCTED, SUPPORTED AND FASTENED AS RECOMMENDED BY SMACNA.

3 DUCTWORK RADIUS ELBOW

SCALE: N.T.S.

FULLY SPRINKLERED
100% CONSTRUCTION DOCUMENTS

Revisions:		Date	CONSULTANTS:		PROJECT MANAGER		Office of Construction and Facilities Management		Drawing Title		Project Title		VA PROJECT NUMBER	
			rpadesign		Headquarters		VA		MECHANICAL DETAILS		RENOVATE/EXPAND		580-17-101	
			5960 Fairview Road, Suite 500		Apogee Consulting Group, P.A.		U.S. Department of Veterans Affairs		2002 HOLCOMBE BOULEVARD,		EMERGENCY ROOM SYSTEM		Building Number	
			Charlotte, NC 28210		1151 Kildaire Farm Road, Suite 120				HOUSTON, TX 77030 4298		Approved: Mark Landwert - COR		# 1	
			704-887-3500		Cary, North Carolina 27511						Date		Drawing Number	
											MARCH 15, 2019		MH501	
											Checked		Dwg 10 of 12	
											SPG			
											WRS			

ISOLATION ROOM CONTROL

GENERAL: THE SUPPLY AIR VAV BOX AND REHEAT COIL SHALL BE CONTROLLED AS SEQUENCED ON MH502.

SPACE PRESSURE CONTROL:

THE DDC SYSTEM SHALL MODULATE THE EXHAUST AIR VENTURI VALVE TO MAINTAIN A SET POINT OF +0.01 INCHES WATER GAUGE TO THE CORRIDOR. OPENING ON A FALL BELOW +0.01 IN FOR MORE THAN 5 SECONDS AND CLOSING ON A RISE ABOVE +0.01 FOR MORE THAN 5 SECONDS.

ALARMS SHALL BE PROVIDED AS FOLLOWS:

- LOW DIFFERENTIAL PRESSURE: IF THE PRESSURE FALLS BELOW 0.00 INCHES FOR MORE THAN 15 SECONDS. A LOCAL VISUAL ALARM SHALL BE PROVIDED WITH A VISUAL MANGHELIC GAUGE AS WELL.

VISUAL INDICATOR:

PROVIDE BALL-THRU-WALL MANUAL PRESSURE DIFFERENTIAL INDICATOR BETWEEN THE ANTE ROOM AND THE CORRIDOR.

FILTERS:

THE DDC SYSTEM SHALL MONITOR THE DIFFERENTIAL PRESSURE ACROSS THE HEPA PRE-FILTER AND THE HEPA FILTER. THE HEPA FILTER SHALL BE PROVIDED WITH A VISUAL MANGHELIC GAUGE AS WELL.

ALARMS SHALL BE PROVIDED AS FOLLOWS:

- WHEN THE PRE-FILTER DIFFERENTIAL PRESSURE EXCEEDS 1.0 INCHES OF WATER GAUGE.
- WHEN THE HEPA FILTER DIFFERENTIAL PRESSURE EXCEEDS 2.0 INCHES OF WATER GAUGE.

FAN CONTROL:

THE TWO FANS SHALL OPERATE IN A LEAD / STANDBY FASHION.

- THE LEAD FAN SHALL RUN FIRST AND THE STANDBY FAN ISOLATION DAMPER SHALL BE CLOSED.
- ON FAILURE OF THE LEAD FAN, THE STANDBY FAN SHALL RUN AND THE LEAD FAN ISOLATION DAMPER SHALL BE CLOSED.

THE DESIGNATED LEAD FAN SHALL ROTATE UPON ONE OF THE FOLLOWING CONDITIONS (USER SELECTABLE):

- MANUALLY THROUGH SOFTWARE SWITCH
- IF FAN RUN TIME (ADJUSTABLE) IS EXCEEDED
- WEEKLY
- MONTHLY

UPON A FALL IN DUCT STATIC PRESSURE AHEAD OF THE FILTER AT A SET POINT AS DETERMINED DURING TAB, THE VFD SHALL MODULATE UP. ON A RISE THE VFD SHALL MODULATE DOWN.

ALARMS TO BOTH FANS SHALL BE PROVIDED AS FOLLOWS:

- FAILURE: COMMANDED ON, BUT THE STATUS IS OFF.
- RUNNING IN HAND: COMMANDED OFF, BUT THE STATUS IS ON.
- RUNTIME EXCEEDED: STATUS RUN TIME EXCEEDS A USER DEFINABLE LIMIT.

4 ISOLATION ROOM CONTROL SEQUENCE OF OPERATION

SCALE: N.T.S.

VAV TERMINAL UNIT

RUN CONDITIONS - CONTINUOUS:

THE UNIT SHALL RUN CONTINUOUSLY AND SHALL MAINTAIN:

- A 75°F (ADJ.) COOLING SETPOINT.
- A 70°F (ADJ.) HEATING SETPOINT.

ALARMS SHALL BE PROVIDED AS FOLLOWS:

- HIGH ZONE TEMP: IF THE ZONE TEMPERATURE IS GREATER THAN THE COOLING SETPOINT BY A USER DEFINABLE AMOUNT (ADJ.).
- LOW ZONE TEMP: IF THE ZONE TEMPERATURE IS LESS THAN THE HEATING SETPOINT BY A USER DEFINABLE AMOUNT (ADJ.).

ZONE SETPOINT ADJUST:

THE OCCUPANT SHALL BE ABLE TO ADJUST THE ZONE TEMPERATURE HEATING AND COOLING SETPOINTS AT THE ZONE SENSOR.

REVERSING VARIABLE VOLUME TERMINAL UNIT - FLOW CONTROL:

THE UNIT SHALL MAINTAIN ZONE SETPOINTS BY CONTROLLING THE AIRFLOW THROUGH ONE OF THE FOLLOWING:

OCCUPIED:

- WHEN ZONE TEMPERATURE IS GREATER THAN ITS COOLING SETPOINT, THE ZONE DAMPER SHALL MODULATE BETWEEN THE MINIMUM OCCUPIED AIRFLOW (ADJ.) AND THE MAXIMUM COOLING AIRFLOW (ADJ.) UNTIL THE ZONE IS SATISFIED.
- WHEN THE ZONE TEMPERATURE IS BETWEEN THE COOLING SETPOINT AND THE HEATING SETPOINT, THE ZONE DAMPER SHALL MAINTAIN THE MINIMUM REQUIRED ZONE AIRFLOW (ADJ.).
- WHEN ZONE TEMPERATURE IS LESS THAN ITS HEATING SETPOINT, THE CONTROLLER SHALL MODULATE THE HEATING VALVE OPEN TO MAINTAIN THE ZONE TEMPERATURE AT ITS HEATING SETPOINT. ADDITIONALLY, IF WARM AIR IS AVAILABLE FROM THE AHU, THE ZONE DAMPER SHALL MODULATE BETWEEN THE MINIMUM OCCUPIED AIRFLOW (ADJ.) AND THE MAXIMUM HEATING AIRFLOW (ADJ.) UNTIL THE ZONE IS SATISFIED.

UNOCCUPIED:

- WHEN THE ZONE IS UNOCCUPIED THE ZONE DAMPER SHALL CONTROL TO ITS MINIMUM UNOCCUPIED AIRFLOW (ADJ.).
- WHEN THE ZONE TEMPERATURE IS GREATER THAN ITS COOLING SETPOINT, THE ZONE DAMPER SHALL MODULATE BETWEEN THE MINIMUM UNOCCUPIED AIRFLOW (ADJ.) AND THE MAXIMUM COOLING AIRFLOW (ADJ.) UNTIL THE ZONE IS SATISFIED.
- WHEN ZONE TEMPERATURE IS LESS THAN ITS UNOCCUPIED HEATING SETPOINT, THE CONTROLLER SHALL MODULATE THE HEATING VALVE OPEN TO MAINTAIN THE ZONE TEMPERATURE AT THE SETPOINT. ADDITIONALLY, IF WARM AIR IS AVAILABLE FROM THE AHU, THE ZONE DAMPER SHALL MODULATE BETWEEN THE MINIMUM UNOCCUPIED AIRFLOW (ADJ.) AND THE AUXILIARY HEATING AIRFLOW (ADJ.) UNTIL THE ZONE IS SATISFIED.

REHEATING COIL VALVE:

THE CONTROLLER SHALL MEASURE THE ZONE TEMPERATURE AND MODULATE THE REHEATING COIL VALVE OPEN ON DROPPING ZONE TEMPERATURE TO MAINTAIN ITS HEATING SETPOINT.

REHEATING - HIGH DISCHARGE AIR TEMPERATURE LIMIT:

THE CONTROLLER SHALL MEASURE THE DISCHARGE AIR TEMPERATURE AND LIMIT HOT WATER VALVE POSITION IF THE DISCHARGE AIR TEMPERATURE IS MORE THAN 15°F (ADJ.) ABOVE THE ZONE TEMPERATURE.

DISCHARGE AIR TEMPERATURE:

THE CONTROLLER SHALL MONITOR THE DISCHARGE AIR TEMPERATURE.

ALARMS SHALL BE PROVIDED AS FOLLOWS:

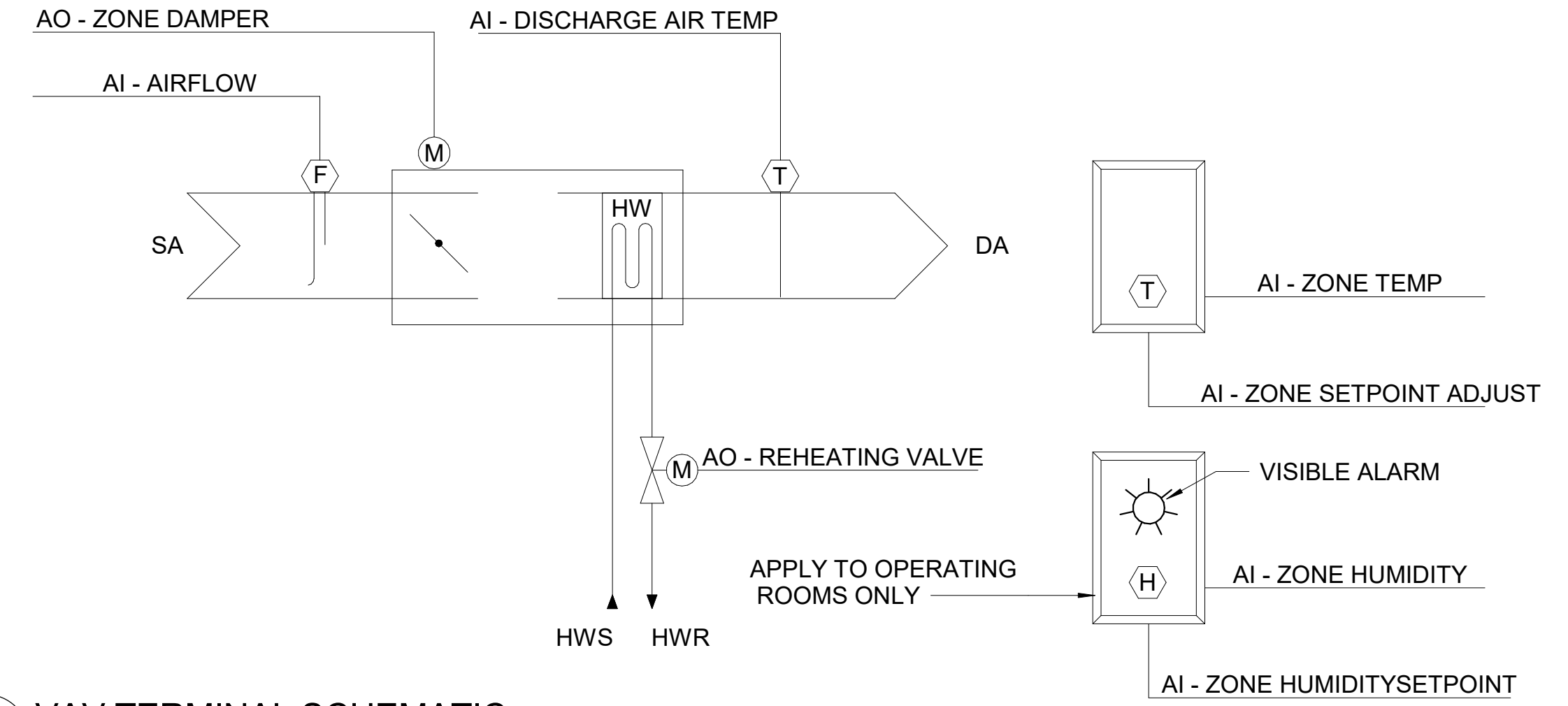
- HIGH DISCHARGE AIR TEMP: IF THE DISCHARGE AIR TEMPERATURE IS GREATER THAN 100°F (ADJ.).
- LOW DISCHARGE AIR TEMP: IF THE DISCHARGE AIR TEMPERATURE IS LESS THAN 50°F (ADJ.).

ALARMS SHALL BE PROVIDED AS FOLLOWS:

- HIGH HUMIDITY: IF THE SPACE RELATIVE HUMIDITY RISES ABOVE 60% A VISIBLE ALARM SHALL BE INITIATED IN THE AFFECTED OR AND AT THE DDC MONITORING STATION.

3 VAV TERMINAL SEQUENCE OF OPERATION

SCALE: 12" = 1'-0"



1 VAV TERMINAL SCHEMATIC

SCALE: 12" = 1'-0"

POINT NAME:	HARDWARE POINTS					SOFTWARE POINTS					SHOW ON GRAPHIC
	AI	AO	BI	BO	AV	BV	LOOP	SCHED	TREND	ALARM	
ZONE TEMPERATURE	X								X		X
ZONE TEMPERATURE SETPOINT											X
AIRFLOW	X								X		X
DISCHARGE AIR TEMPERATURE	X								X		X
ZONE DAMPER		X									X
REHEATING VALVE		X							X		X
MAXIMUM OCCUPIED AIRFLOW SETPOINT					X				X		X
HEATING MODE						X			X		
HEATING SETPOINT									X		X
COOLING SETPOINT									X		X
HIGH ZONE TEMPERATURE										X	
LOW ZONE TEMPERATURE										X	
HIGH DISCHARGE TEMPERATURE										X	
LOW DISCHARGE TEMPERATURE										X	
HIGH ZONE HUMIDITY										X	
MINIMUM OCCUPIED AIR FLOW SETPOINT					X				X		X
MINIMUM UNOCCUPIED AIRFLOW SETPOINT					X				X		X
ZONE HUMIDITY	X								X		
ZONE HUMIDITY SETPOINT	X								X		

